

# Can Quantum Chemistry Referee Metabolic $\Delta_r G'^{\circ}$ ?

A QM/ML composite vs. dGPredictor on TECRDB disagreements

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ModelSEED thermodynamics project

Progress summary — July 2026

# The question, and the short answer

## Motivating question

ModelSEED's retrained **dGPredictor** disagrees with experiment (TECRDB) on a set of reactions by 16–100 kJ/mol. Can a **parameter-free quantum-mechanical** method *adjudicate* those disagreements — from physics, sharing none of dGPredictor's group-contribution basis?

**Short answer: no — and it need not.**

- The data-driven **eQuilibrator** already matches experiment to within its noise (MAE **3.6** kJ/mol, exp. sd 6.2).
- Our QM composite sits at **38.3** kJ/mol — it cannot referee a 100 kJ/mol gap when its own error is  $\sim 38$ .

**What QM *can* defensibly do**

- On glutathione redox its *magnitude* ( $\sim 36$  kJ/mol) independently excludes dGPredictor's  $\sim 103$  —
- from a method trained on **zero** thermodynamic data.
- That corroboration, not a competitive MAE, is the usable result.

# Methodology

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# The composite — what is computed

For every metabolite we build an aqueous standard Gibbs energy, then combine species into reactions and transform to physiological conditions.

$$G_{\text{aq}} = \underbrace{E_{\text{MLIP}}^{\text{gas}}}_{\text{UMA / MACE-POLAR}} + \underbrace{\Delta G_{\text{solv}}^{\text{ALPB}}}_{\text{xtb GFN2}} + \underbrace{G_{\text{RRHO}}}_{\text{quasi-RRHO thermal}}$$

1. Enumerate conformers; optimise **in ALPB water**.
2. Single-point gas energy on that same geometry (electronic + solvation share one structure).
3. One Hessian per compound → thermal correction.
4. **Boltzmann-average** over conformers.
5. Combine into  $\Delta_r G'^{\circ}$  per reaction.
6. **Alberty** Legendre transform to the measured pH + extended **Debye–Hückel** (ionic strength).

## Key discipline

Corrections are calibrated *only* on external experimental **pKa** values — 2/11

## Why this design, and what it is *not*

### The error budget (per reaction)

A solvated polyanion computed absolutely carries continuum-solvation error on ions ( $\sim 15$  kJ/mol/species), plus conformational, RRHO and electronic error. Four species, uncorrelated:

$$\sqrt{4} \times 15 \approx 30 \text{ kJ/mol}$$

Measured scatter: 42 (130 rxns), 34 (cleanest subset). The gap to the 3–8 kJ/mol “useful” target is **structural**, not a tuning problem — and reproduces Jinich *et al.*, *Sci. Rep.* 2014.

### The bottleneck is *not* the electronic method

- Substituting real DFT (r2SCAN-3c) for the MLIP at fixed geometry made it **worse** (14.9  $\rightarrow$  17.3).
- UMA vs. MACE-POLAR agree at  $r = 0.98$  (spread 7 vs. error 40).
- Solvation is the driver: continuum models **under-solvate multiply-charged anions** ( $\sim 30$  kJ/mol per reaction).

$\Rightarrow$  every phosphate/polyanion pays the same structural tax.

# Implementation

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## Pipeline — data in, $\Delta_r G^\circ$ out

Nothing is hard-coded to a reaction set: reactions, metabolites and measurement conditions are all read from data.

| #   | stage            | where            | what                                             |
|-----|------------------|------------------|--------------------------------------------------|
| 1   | build_inputs     | CPU              | ModelSEED reactions/metabolites → JSON           |
| 2   | microspecies     | CPU              | audit SMILES for protonation / hydration state   |
| 3–4 | build_ensembles  | CPU (xtb)        | ETKDG embed → GFN2 opt (ALPB) → Hessian          |
| 5   | run_uma_ensemble | <b>multi-GPU</b> | UMA gas energy per conformer, Boltzmann avg.     |
| 6   | analyze_pka      | CPU+GPU          | anion-solvation calibration (one-time, reusable) |
| 7   | final_model      | CPU              | score reactions → perreaction_dG.csv             |
| 8   | plot_comparison  | CPU              | figures                                          |

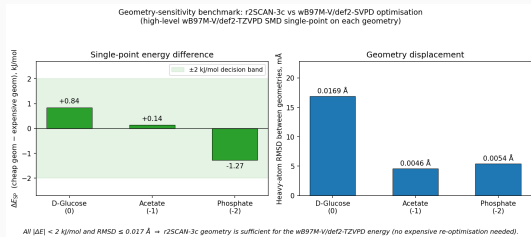
**Compute** (23 metabolites, 605 conformers):  
conformer ensembles ~35 min (dominates, ~2  
core-h/metabolite); UMA scoring ~3.5 min on  
8×V100.

**Scales linearly:** ~1000 metabolites ≈ 2000  
core-hours ≈ 25 h on 80 cores; GPU time  
negligible.

# Two engineering checks that matter

## Geometry sufficiency

r2SCAN-3c geometry vs. expensive wB97M-V/def2-TZVPD single point: all  $|\Delta E| < 2 \text{ kJ/mol}$  and  $\text{RMSD} \leq 0.017 \text{ \AA}$ .  
*Cheap geometries are good enough* — no expensive re-optimisation.



## The Alberty transform passes independently

- ATP hydrolysis  $-35.3$  vs. published  $-32.5$ .
- Debye-Hückel scales as  $\Delta(z^2)$  exactly 1:4:9:16.
- pH slope exact.

## Bugs found & fixed (each moved the answer):

- Missing standard-state term  
 $RT \ln(24.46) = 7.93 \text{ kJ/mol/species}$  (hid in  $\Delta n = 0$  isomerisations).
- Quality screen rejected acetic acid — now screens on imaginary-mode *magnitude*



## Results

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## Headline — the ten disagreement reactions

| method               | MAE vs. TECRDB           |
|----------------------|--------------------------|
| <b>eQuilibrator</b>  | <b>3.6</b> (exp. sd 6.2) |
| predict zero         | 10.5                     |
| QM, isodesmic (best) | 24.6                     |
| GroupContribution    | 35.4                     |
| <b>QM composite</b>  | <b>38.3</b>              |
| <b>dGPredictor</b>   | <b>61.2</b>              |

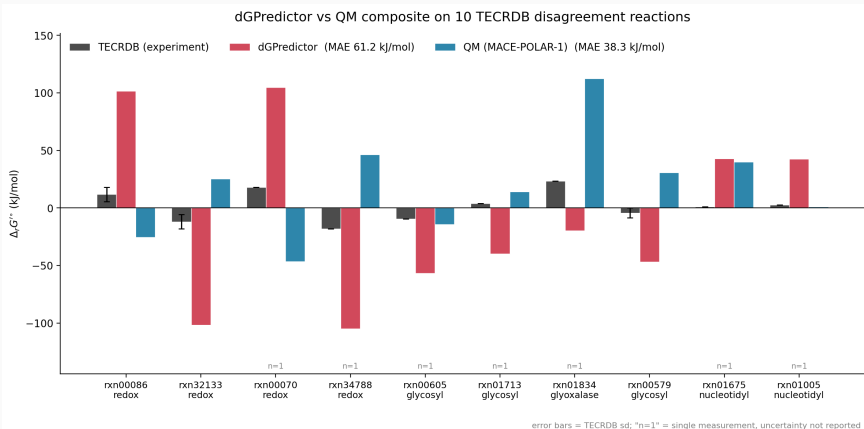
### The ten are already solved

eQuilibrator agrees with TECRDB on *exactly* the reactions dGPredictor fails. So the adjudication verdict is “**dGPredictor is the outlier**” — and it needs no quantum chemistry.

Database-wide (1550 TECRDB $\leftrightarrow$ ModelSEED matched reactions): eQuilibrator covers 84% at MAE 5.5; dGPredictor covers 100% at 13.2.

*Caveat:* the 10 rows are 2 redox chemistries written both directions + 6 single-measurement reactions  $\approx$  8 independent points.

## Per-reaction $\Delta_r G'^{\circ}$ — the ten selected reactions



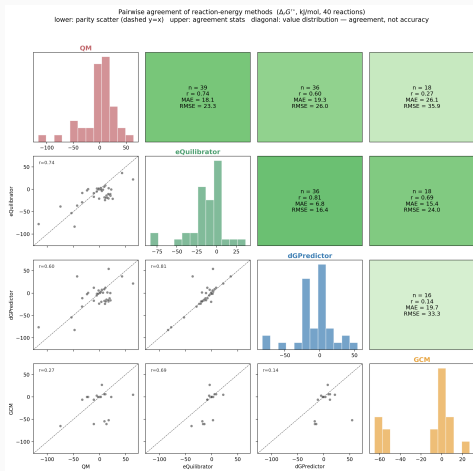
dGPredictor (red) swings to  $\pm 100$  kJ/mol on the redox reactions; QM (blue) is closer but wrong-signed on several. Neither tracks the small experimental values (grey).

## Per-reaction $\Delta_r G'^{\circ}$ — the numbers

| #                       | reaction                                              | class       | TECRDB | dGP    | QM    |
|-------------------------|-------------------------------------------------------|-------------|--------|--------|-------|
| 1                       | rxn00086 glutathione:NADP <sup>+</sup> oxidoreductase | redox       | 11.9   | 101.6  | -25.3 |
| 2                       | rxn32133 glutathione reductase (NADPH)                | redox       | -11.9  | -101.6 | 25.3  |
| 3                       | rxn00070 glutathione:NAD <sup>+</sup> oxidoreductase  | redox       | 18.0   | 104.9  | -46.5 |
| 4                       | rxn34788 glutathione-disulfide reductase (chlpl.)     | redox       | -18.0  | -104.9 | 46.5  |
| 5                       | rxn00605 UDP-glucose:G6P glucosyltransferase          | glycosyl    | -9.5   | -56.8  | -14.3 |
| 6                       | rxn01713 UDP-glucose:sinapate glucosyltransferase     | glycosyl    | 3.9    | -39.8  | 14.2  |
| 7                       | rxn01834 (R)-S-lactoylglutathione lyase               | glyoxalase  | 23.5   | -19.8  | 112.6 |
| 8                       | rxn00579 UDP-glucose:fructose glucosyltransferase     | glycosyl    | -4.2   | -46.9  | 30.8  |
| 9                       | rxn01675 dTTP:glucose-1-P thymidyltransferase         | nucleotidyl | 1.0    | 42.9   | 39.9  |
| 10                      | rxn01005 UTP:phospho-glucuronate uridylyltransferase  | nucleotidyl | 2.7    | 42.7   | 1.2   |
| MAE vs. TECRDB (kJ/mol) |                                                       |             | —      | 61.2   | 38.3  |

All values  $\Delta_r G'^{\circ}$  in kJ/mol, pH-matched. dGPredictor's  $\pm 100$  kJ/mol redox swing (rows 1–4) is the failure QM was asked to referee; QM's magnitude ( $\sim 36$ ) sits with experiment (15), not with the  $\pm 100$  group-contribution regime — the corroboration that matters.

# Where the four methods agree, and where they don't



Pairwise agreement over 40 reactions (agreement, *not* accuracy):

- eQuilibrator ↔ dGPredictor agree best ( $r = 0.81$ , MAE 6.8) — same data lineage.
- QM correlates with all three ( $r = 0.6$ – $0.74$ ) but with a wide MAE ( $\sim 18$ – $26$ ) — an *independent* axis.
- GCM (GroupContribution) is the weakest ( $r \leq 0.27$  vs. QM), and it is what dGPredictor shares a basis with.

# The decisive negative result — why no correction rescues it

## Per-species anchoring is dead

Compound-grouped cross-validation on per-species offsets:

$$R^2 = 0.177 \quad (\text{in-sample } 0.769)$$

To tie predict-zero needs  $R^2 \approx 0.89$ ; to match eQuilibrator  $\approx 0.97$ . This rules out every anchoring scheme at once ( $\mu_H$ , cofactor  $E'^\circ$ , anion-site corrections, reference-compound reformulation). The error is not in that subspace.

## Mechanisms eliminated

| mechanism                   | verdict             |
|-----------------------------|---------------------|
| electronic structure        | ✗ (DFT worse)       |
| MLIP architecture           | ✗ ( $r = 0.98$ )    |
| conformer sampling          | ✗ (wrong sign)      |
| Alberty transform           | ✓ passes            |
| bond-graph artifact         | ✗ (0 changed)       |
| protonation state           | ✓ correct           |
| Mg <sup>2+</sup> speciation | ✗ (+0.3 kJ/mol net) |

Diagnosis is **solid**; every candidate correction was **rejected** on its own test.

**Where this leaves us**

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# Summary & where QM should be pointed

## What we established

- eQuilibrator solves the TECRDB disagreements; QM cannot beat it on aqueous polyanion chemistry.
- The limit is **continuum anion solvation** ( $\sim 30$  kJ/mol/reaction) — structural, reproduces the literature.
- Robust across electronic method, MLIP, CV, solvation swap and  $\text{Mg}^{2+}$  bookkeeping.
- **The one real QM win:** on glutathione redox its magnitude independently excludes dGPredictor's  $\pm 100$  from zero training data

## The open problem

- QM is uniquely positioned where data-driven methods *cannot* score: the 16% of matched reactions eQuilibrator misses, stereochemistry (GCM is blind by construction), novel metabolites.
- But that regime has **no validation data by definition**.
- Next: design held-out compound classes / stereoisomer pairs *before* more compute; long-term ceiling is explicit-solvent MD + MLIP + alchemical free energy.



The physics cannot reach 5 kJ/mol on polyanions —  
but on the reactions that matter, it agrees with experiment  
*for reasons the data-driven methods cannot claim.*